

PROGRAMMING FUNDAMENTALS

Spring 2024

22nd Feb

Nested IF statement

Decision Table

- A matrix representation of the logic of Nested - IFs
- Lists causes and effects in a matrix format
- Specifies the possible conditions and the resulting actions
- Each column represents a unique combination of conditions
- Best used for complicated decision logic

Parts of Decision Table

- Condition Stubs → Listed conditions
- Action Stubs → Actions which need to be carried out
- Rules → Specify which actions will be carried out for a given set of conditions
- Total number of Rules = 2^n (where n = no. of conditions)

Conditions	Condition Alternatives
Actions	Action Entries

General format

Causes		Values	1	2	3	4	5	6	7	8
Cause 1		Y, N	Y	Y	Y	Y	N	N	N	N
Cause 2		Y, N	Y	Y	N	N	Y	Y	N	N
Cause 3		Y, N	Y	N	Y	N	Y	N	Y	N
Effects										
Effect 1			X			X				X
Effect 2				X				X		X

How to construct a decision table?

- Name the conditions in the condition stub area
- Name all possible actions that can occur
- List all rules
- Define the actions for each rule

Decision Tree- Ex-1

Conditions	Codes	R1	R2	R3	R4	R5	R6	R7	R8
Requested a Catalog?	C1	Y	Y	Y	Y	N	N	N	N
Ordered in past two years?	C2	Y	Y	N	N	Y	Y	N	N
Signed on Web site?	C3	Y	N	Y	N	Y	N	Y	N
Actions									
Remain on mailing list	A1	X	X	X	X	X	X	X	-
Receive Certificate	A2	X	-	X	-	X	-	X	-

Solution Ex-1

```
IF C1 THEN
  IF C2 THEN
    IF C3 THEN
      DO A1, A2
    ELSE
      DO A1
    END IF
  ELSE
    IF C3 THEN
      DO A1, A2
    ELSE
      DO A1
    END IF
  END IF
ELSE
  IF C2 THEN
    IF C3 THEN
      DO A1, A2
    ELSE
      DO A1
    END IF
  ELSE
    IF C3 THEN
      DO A1, A2
    END IF
  END IF
END IF
```


Exercise-2

Conditions	Codes	R1	R2	R3	R4	R5	R6	R7	R8
Normal Month?	C1	Y	Y	Y	Y	N	N	N	N
Festive Month?	C2	Y	Y	N	N	Y	Y	N	N
Unsold?	C3	Y	N	Y	N	Y	N	Y	N
Actions									
Order at 2% projected sales	A1	-	-	X	X	-	-	-	-
Order at 4-8% projected sales	A2	-	-	-	-	X	X	-	-
Adjust order for un-sold	A3	-	-	X	-	X	-	-	-
Offer Clearance sale at special outlets	A4	-	-	X	-	-	-	X	-
Offer festive sale at alloutlets	A5	-	-	-	-	X	X	-	-
Carry out regular operations	A6	X	-	-	-	X	X	-	-

Solution-2

```
IF C1 THEN
    IF C2 THEN
        IF C3 THEN
            DO A6
        END IF
    ELSE
        IF C3 THEN
            DO A1, A3, A4
        ELSE
            DO A1
        END IF
    END IF
ELSE
    IF C2 THEN
        IF C3 THEN
            DO A2, A3, A5, A6
        ELSE
            DO A2, A5, A6
        END IF
    ELSE
        IF C3 THEN
            DO A4
        END IF
    END IF
END IF
```

Ex-3

Conditions	Codes	R1	R2	R3	R4
Book Availability?	C1	Y	Y	N	N
Book Classification?	C2	Y	N	Y	N
Actions					
Reserve the book	A1	-	-	X	-
Reject Borrowing	A2	X	-	-	-
Stamp Due Date	A3	-	X	-	-
Issue Book	A4	-	X	-	-

Solution Ex-3

```
IF C1 THEN
    IF C2 THEN
        DO A2
    ELSE
        DO A3, A4
    END IF
ELSE
    IF C3 THEN
        DO A1
    END IF
END IF
```

Application areas

- Business Analysis
- Programming
- Testing
- Hardware Design

Question-4 Let's Practise

Conditions	Codes	R1	R2	R3	R4	R5	R6	R7	R8
First Condition?	C1								
Second Condition?	C2								
Third Condition?	C3								
Actions									
1st Action	A1	-	-	X	X	-	-	-	-
2nd Action	A2	X	-	-	X	X	X	X	-
3rd Action	A3	-	X	X	-	X	-	-	-
4th Action	A4	X	-	-	-	X	X	-	X

Convert into Nested-IF structure after completing the table on your sheet
Submit before the end of class

Combining AND and OR operator

- Most programming languages allow you to combine as many AND and OR operators in an expression
- When you combine AND and OR operators, the AND operators take precedence

Q1

Lets assume that your program is meant to display factorial of any number from 2 to 10. Ask the user to enter an integer value. Using Switch case structure, display factorial of each value from 2 to 10 according to table:

2	2
3	6
4	24
5	120
6	720
7	5040
8	40320
9	362880
10	328800

 D:\Books-ICT\C-Programs\factorial_using_Switch.exe

Enter a number : 5

120

Process exited after 4.439 seconds with return code 0

Press any key to continue . . .

 D:\Books-ICT\C-Programs\factorial_using_Switch.exe

Enter a number : 12

Invalid Input

Process exited after 3.337 seconds with return code 0

Press any key to continue . . .

Book Reference

- D.S. Malik (ch - 4)
- Tony Gaddis (ch - 4)